

LAB MANUAL 1

CLOUD COMPUTING

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BSSE 8th Morning

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Title: Simulation of Cloud Resource Allocation using CloudSim Toolkit in Eclipse IDE

1. Purpose

The purpose of this lab is to set up a proper development environment by installing Java and Eclipse IDE, then integrating the CloudSim toolkit into the project. After setup, a sample simulation program is executed to understand how cloud components like virtual machines and tasks are managed. This experiment mainly focuses on learning how cloud resources are distributed and utilized efficiently.

2. Background Overview

Cloud computing refers to delivering computing services such as storage, servers, and applications through the internet instead of using local systems. It allows flexible and scalable resource usage without physical hardware dependency.

In this practical, a built-in example from CloudSim is executed to observe how a task (cloudlet) is assigned to a virtual machine and processed. This helps in understanding the working of a simulated cloud environment.

3. Tools and Technologies Used

The following tools were required to complete this lab:

1. Java Development Kit (JDK 8 or above)

Java Development Kit is necessary for writing and executing Java programs. It provides essential tools like compiler and runtime environment.

Download link:

<https://www.oracle.com/java/technologies/javase-downloads.html>

The screenshot shows the Oracle Java Downloads page for JDK 26. The browser tabs include 'Java Downloads | Oracle'. The page content states: 'JDK 26 binaries are free to use in production and free to redistribute, at no cost, under the Oracle No-Fee Terms and Conditions (NFTC). JDK 26 will receive updates under these terms, until September 2026, when it will be superseded by JDK 27.' Below this, there are tabs for 'Linux', 'macOS', and 'Windows'. The 'Linux' tab is active, displaying a table of download links for various Linux architectures.

Product/file description	File size	Download
ARM64 Compressed Archive	208.96 MB	https://download.oracle.com/java/26/latest/jdk-26_linux-aarch64_bin.tar.gz (sha256)
ARM64 RPM Package	208.53 MB	https://download.oracle.com/java/26/latest/jdk-26_linux-aarch64_bin.rpm (sha256) (OL 9 GPG Key)
x64 Compressed Archive	211.05 MB	https://download.oracle.com/java/26/latest/jdk-26_linux-x64_bin.tar.gz (sha256)
x64 Debian Package	180.82 MB	https://download.oracle.com/java/26/latest/jdk-26_linux-x64_bin.deb (sha256)
x64 RPM Package	210.58 MB	https://download.oracle.com/java/26/latest/jdk-26_linux-x64_bin.rpm (sha256) (OL 9 GPG Key)

Below the table is a 'Documentation Download' button. At the bottom, there is a 'Release information' section.

2. Eclipse IDE for Java Developers

Eclipse is used as the coding platform where projects are created and executed. It provides features for writing, debugging, and managing Java applications.

Download link:

<https://www.eclipse.org/downloads/>

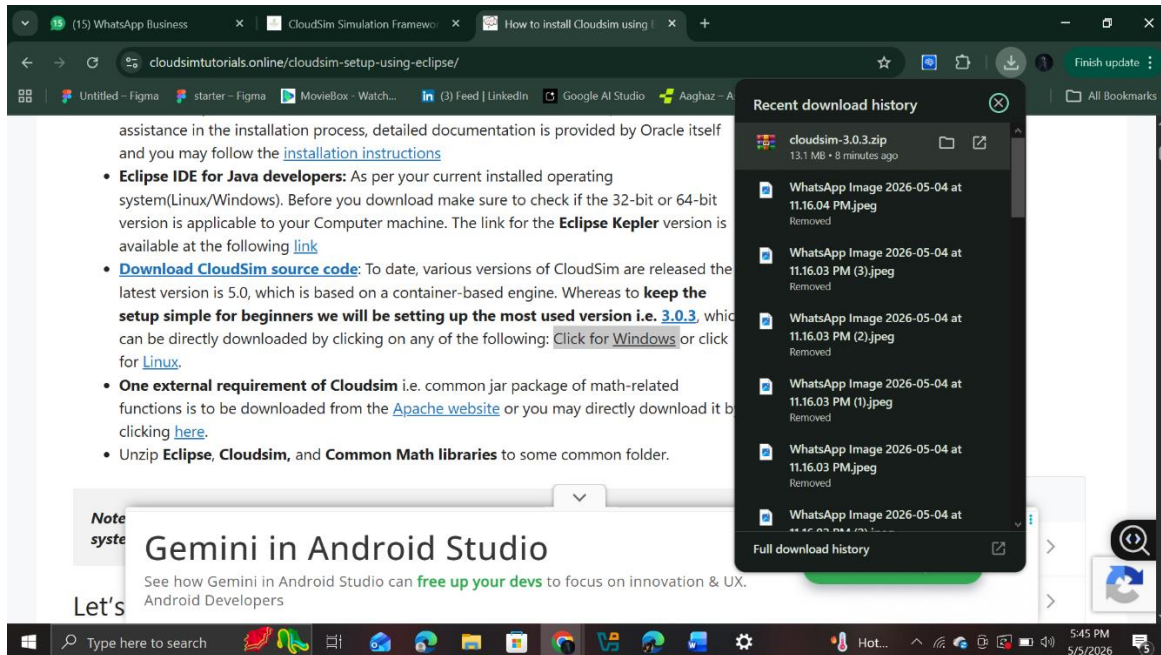
The screenshot shows the Eclipse IDE for Java Developers download page. The browser tabs include 'Eclipse IDE for Java Developers'. The page features the Eclipse Foundation logo and navigation links: 'Projects', 'Supporters', 'Collaborations', 'Resources', and 'The Foundation'. A 'Download' button is visible. Below the navigation, there are tabs for 'Installer', 'Eclipse Packages', and 'Eclipse Developer Builds'. The 'Eclipse Packages' tab is active, showing the path: 'Home > Downloads > Release > Packages > Kepler > SR1 > Eclipse IDE for Java Developers'. The main heading is 'Eclipse IDE for Java Developers'. Below this, there is a 'Package Description' section with an Eclipse logo. To the right, there is a 'Download Links' section with links for 'Windows: 32-bit | x86_64', 'macOS: 32-bit', and 'Linux: 32-bit | x86_64'. Further right, there is a 'RELATED LINKS' section with a link to 'Compare & Combine Packages'. At the bottom, there is a cookie consent banner with 'Decline' and 'Allow cookies' buttons.

3. CloudSim Toolkit

CloudSim is a simulation framework used to model cloud computing environments. It includes libraries and sample programs for experimentation.

Download link:

Click for [Windows](#)



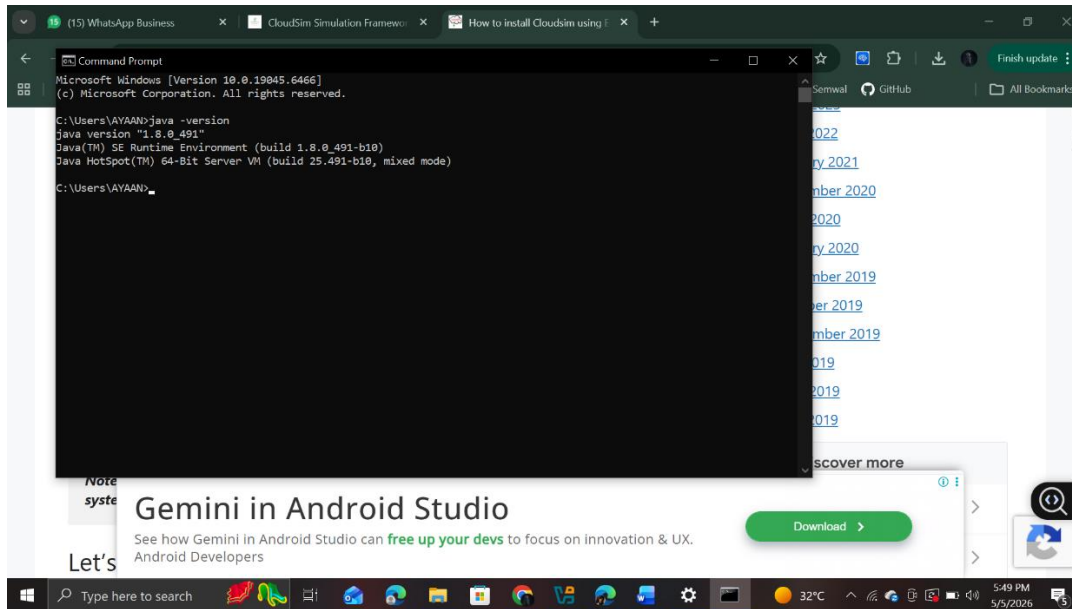
4. Implementation Steps

Step 1: Setting up Java Development Environment

Initially, the Java Development Kit was downloaded and installed on the system. After installation, the command prompt was used to verify whether Java was correctly installed.

Command used:

java -version



The display of version details confirmed successful installation.

Step 2: Installing Development Platform (Eclipse IDE)

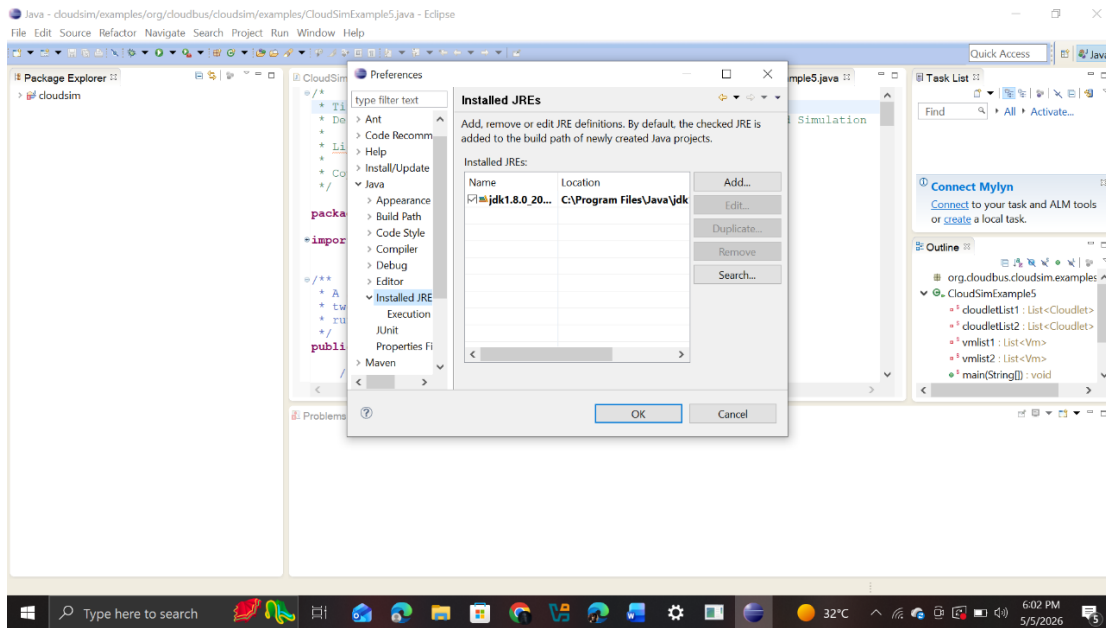
After installing Java, Eclipse IDE was downloaded and launched. A workspace folder was selected to store all project-related files. Eclipse provides an easy interface for coding and execution.

Step 3: Configuring Java Environment in IDE

To ensure Eclipse uses the correct Java version, JDK was configured manually.

Steps followed:

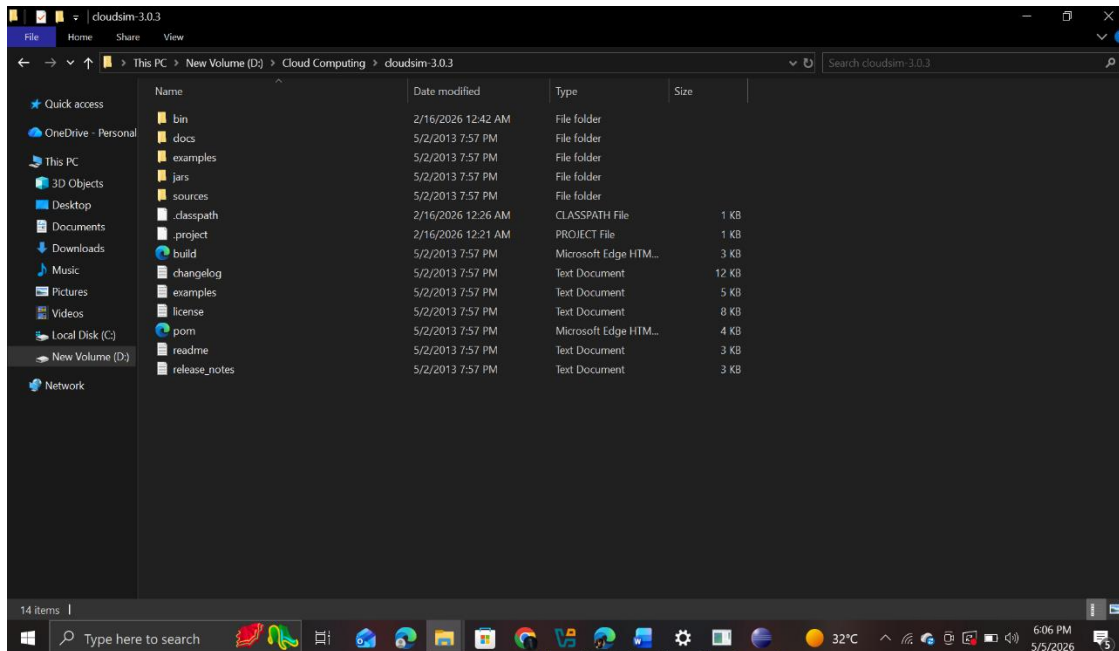
- Open: Window → Preferences → Java → Installed JREs
- Add the installed JDK path
- Set it as the default option



This ensures smooth compilation and execution of programs.

Step 4: Obtaining Cloud Simulation Toolkit

The CloudSim toolkit was downloaded. The compressed file was saved on the system for further setup.



Step 5: Extracting and Structuring Project Files

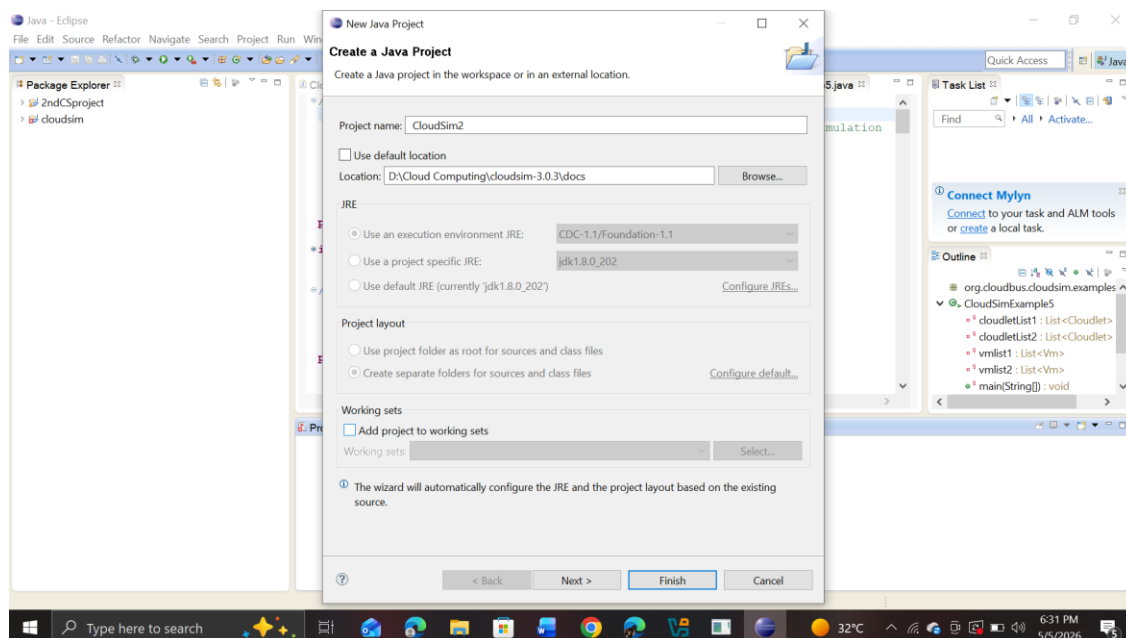
The downloaded ZIP file was extracted. All files including libraries and example programs were arranged in a single folder to keep everything organized and easily accessible.

Step 6: Initializing New Java Project

A new Java project named **CloudSim2** was created inside Eclipse. This project serves as the workspace for running CloudSim simulations.

Steps followed:

- File → New → Java Project
- Enter project name
- Click Finish

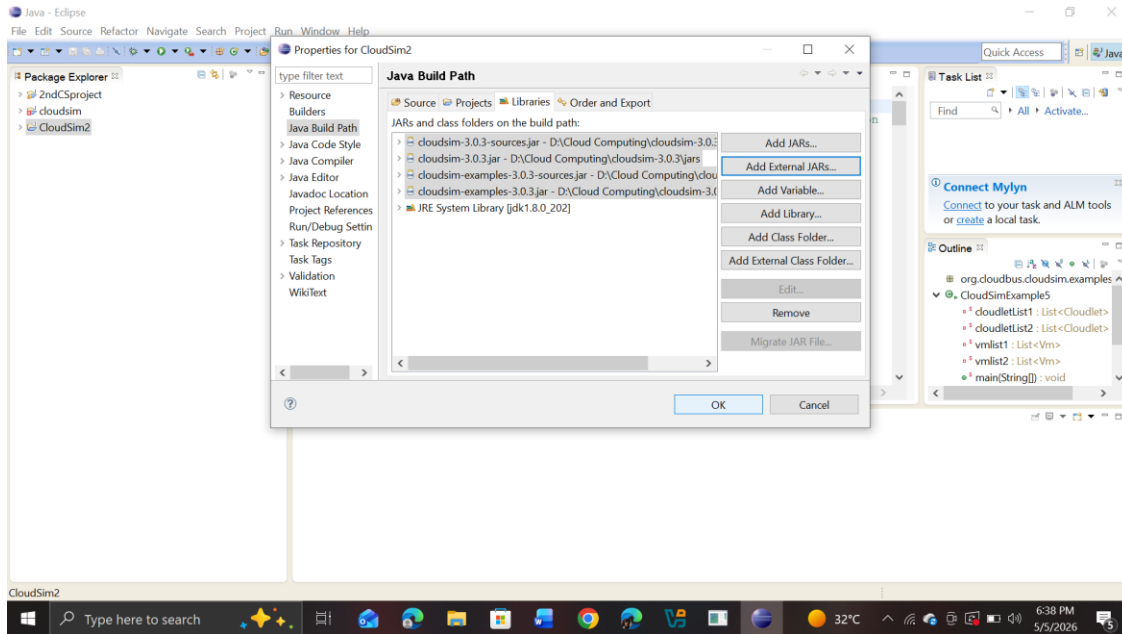


Step 7: Importing Required Libraries

All required JAR files were added to the project so that CloudSim classes could be used.

Steps followed:

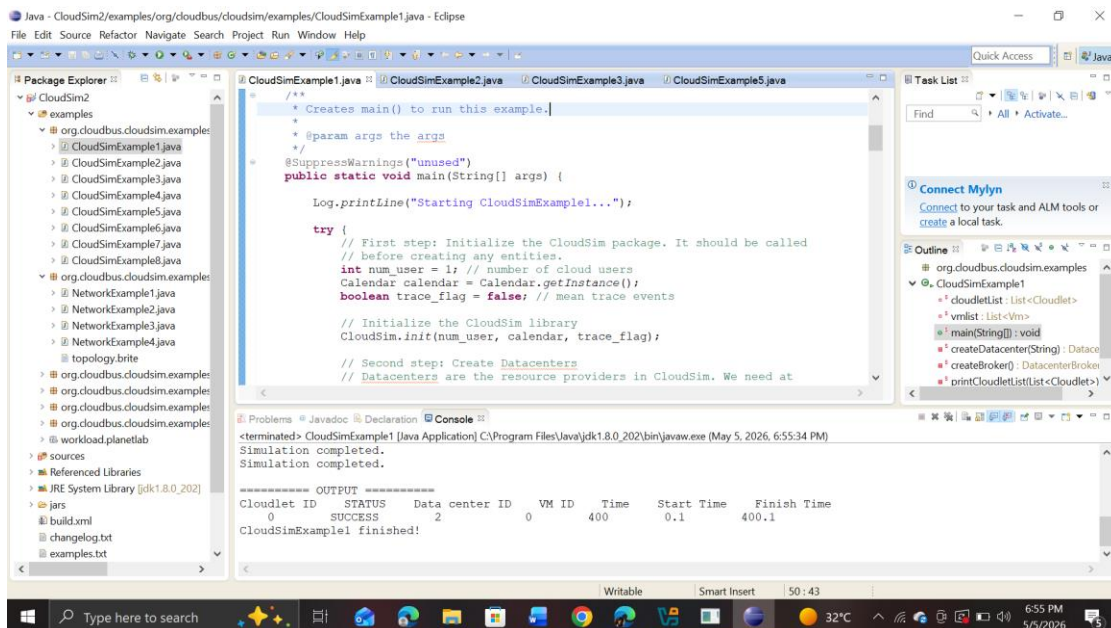
- Right-click project → Build Path → Configure Build Path
- Open Libraries tab
- Click “Add External JARs”
- Select all JAR files from the CloudSim folder



This allows access to simulation classes such as Datacenter, VM, and Broker.

Step 8: Executing Sample Simulation Program (CloudSimExample1)

Instead of creating a new program, a pre-built example file named **CloudSimExample1.java** was used. This file already contains a complete simulation setup, making it easier to understand the working process.



5. Outcome of the Experiment

The simulation program ran successfully without any errors. It demonstrated that the cloudlet was properly allocated to a virtual machine and executed efficiently, confirming correct working of the system.

6. Summary of Findings

This experiment helped in understanding the basic concept of cloud simulation using CloudSim. It showed how different components like datacenters, virtual machines, and tasks interact within a cloud environment. Simulation tools like CloudSim are useful for testing cloud systems without needing real infrastructure.